Yash Bavadiya

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EDUCATION

Indian Institute of Technology, Jodhpur **Aug 2024 – Jul 2028**
Bachelor of Technology - Computer Science and Engineering - **CGPA: 7.21** Jodhpur, Rajasthan

TECHNICAL SKILLS

Languages: C++, Rust, TypeScript, JavaScript, Python, C, SQL
Systems & AI: Distributed Systems, Discrete-Event Simulation, Concurrency, OS, Networks, DBMS, RAG, LLM Agents, MCP
Technologies: Qt, React, Node.js, WebAssembly, PostgreSQL, SQLite, Redis, WebSockets, Docker, Linux, GitHub Actions

EXPERIENCE

Workers IO (YC F26) **Sep 2026 – Present**
Developer Relations Engineer | Deterministic simulation testing Remote (San Francisco)

- Recruited by the founder after building **Breakscale**, an open-source system design simulator; now its core maintainer with **2,000+ GitHub stars**, 168 forks, **1,200+ tests** and 31 merged PRs from **15 outside contributors**.
- Built on a **discrete-event engine** (a time-ordered event queue), so queue buildup, timeouts, retry storms and circuit-breaker trips emerge from simulation instead of formulas, across **33 component types**.
- Shipped **fault injection** (crash, slow, partition, packet loss) with **deterministic replay links**: the same seed and fault log replay the exact failure from one shareable URL.
- Built **breakscale-mcp**, an **MCP server** that lets AI assistants model a codebase as a runnable simulation, plus a **VS Code extension**; Breakscale is in the Vercel, Claude and Mintlify open-source programs.

Google Summer of Code 2026 **May 2026 – Aug 2026**
Kdenlive @KDE Community | Open Source Contributor, C++ and Qt Remote

- Designed and shipped **per-channel RGB curves**, a **multi-stop gradient editor** and **time-remap keyframe types** in **C++ and Qt**, wired into Kdenlive's MLT filter pipeline and merged for **Kdenlive 26.08** (41,000 monthly Flathub installs).
- Fixed an **11-year-old bug** (reported 2015) with an interval-merging **gap-detection algorithm** that handles gaps covered by other tracks; **23 merge requests** merged after core-maintainer review.

Undergraduate Researcher **Jul 2025 – Nov 2025**
Indian Institute of Technology, Jodhpur | Supervisor: Dr. Pratik Mazumder Jodhpur

- Built the retrieval pipeline of an **offline RAG system** over 300-page PDFs: overlapping chunks, **FAISS** vector search and a **4-bit quantized Phi-3**, answering in **∼3 s** on a laptop GPU; benchmarked models and prompts on ROUGE (**+71.77%** over baseline).

PROJECTS

HatchZip | GitHub | Website | **Rust, Slint, ZIP/7z/RAR/tar** **Oct 2026**

- Archive extractor that **frees disk space while it extracts**: it saves the index aside, extracts from the last entry backwards and **truncates the archive** after each batch; for `.tar.gz` it **punches holes** in consumed bytes.
- **Crash-safe write ordering**, proven by tests that kill the process at every write, save and truncate, then resume and diff every file; **130+ tests** on 3 OSes also verify freed space on ext4, APFS and NTFS.
- Extracts a 200 MB zip in **0.3 s vs 1.5 s** for `unzip`; **2,700+ stars**, **27,000+ downloads**, the **#1 most-starred new Rust repo** of its launch week, and **2 security advisories** fixed for outside researchers.

gyotaku | GitHub | Website | **Rust, GPUI, ONNX Runtime, SQLite FTS5** **Oct 2026**

- Offline search over every screenshot by the text inside it: new screenshots run through **PaddleOCR on ONNX Runtime**, and text plus word positions go into a **SQLite FTS5 trigram index**, so matches highlight in place.
- **1–10 ms per keystroke** across 5,000+ screenshots, 0.77 s from save to searchable, **37 MB idle memory**; **390+ stars** and **4,000+ downloads** on macOS, Windows and Linux.

Flock | GitHub | npm | **TypeScript, WebSocket, Redis, React, Docker** **Jun 2026**

- Multiplayer cursors and live presence in a **10-line React integration**; servers scale horizontally through a **Redis pub/sub relay**, with TTL keys and **keyspace notifications** evicting clients that drop without closing.
- Client-side throttling, cursor smoothing that retargets mid-animation and **exponential-backoff reconnects**; 3 npm packages with **1,100+ downloads** and **58 tests**.

Game Boy Emulator | GitHub | **C++20, SDL2, CMake** **Jul 2026**

- Hand-written DMG emulator: CPU with all **501 opcodes**, all 5 interrupts, a DIV-based timer, PPU and **MBC1/MBC3** banking with battery saves; passes all **11 Blargg CPU test ROMs** and **dmg-acid2**, and boots Pokémon Red.

ACHIEVEMENTS

- **Open source:** fixes merged into Deno, Adobe React Spectrum, Kubernetes SIGs, nlohmann/json, Vitest, Nuxt and jj.
- **Building in public: 3,000+ followers** on X, **3.9M impressions** in 3 months; HatchZip's launch post hit **1.9M views**.
- JEE Advanced 2024 AIR **3449** of 180,000+; JEE Main **99.65** percentile; **450+** LeetCode problems; Codeforces max **1429**.
- **Junior Core**, DevlUp Labs: mentored 30 students into open source and ran 2 workshops; **Core Member**, Programming Club and E-Cell.